



CodeNestXAcademy

Project Proposal

Restaurant Ordering System

Group Member

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Course: C++ Programming Language

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1. Introduction

The Restaurant Ordering System is a C++ application designed to simplify the ordering process in a small restaurant. It allows customers to browse menu items, place orders, and receive a bill automatically. The system applies Object-Oriented Programming (OOP) concepts to organize the program into reusable classes. This project improves order accuracy and provides practical experience in software development.

2. Project Objectives

- Develop a restaurant ordering system using C++.
- Apply Object-Oriented Programming concepts such as encapsulation, inheritance, and polymorphism.
- Allow users to view the menu, place orders, and generate bills.
- Create a user-friendly and organized console application.

3. Key Features

- Display food and drink menus.
- Add items to the customer's order.
- Calculate the total bill automatically.
- Remove or update ordered items.
- Display an itemized receipt.
- Save and manage menu information using data structures.

4. Technical Tools

- Programming Language: C++
- Development Environment: Visual Studio Code
- Compiler: GCC (MinGW)
- Concepts Used: Object-Oriented Programming (Classes, Objects, Encapsulation, Inheritance, Polymorphism)
- Data Structures: Linked List, List, and Structures
- Version Control: Git & GitHub

5. Expected Outcomes

- A fully functional console-based Restaurant Ordering System.
- Better understanding of Object-Oriented Programming and data structures.
- Improved C++ programming and problem-solving skills.
- A project that can be further enhanced with file handling, database integration, or a graphical user interface (GUI) for the final project.

6. Team Contribution

- Phum Vitin: Designed the system structure, implemented the menu management module, and assisted with testing and debugging.
- Thoun Soktheara: Developed the ordering and billing modules, integrated the system, and prepared the project documentation and presentation.

7. Additional Information

Restaurant Ordering System – Flowchart

